

CSA1006 – SOFTWARE ENGINEERING

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UNIT – 1 ASSIGNMENT

Problem 1: Applying Scrum in a Telemedicine App Project

Scenario

A company is developing a telemedicine mobile application that allows patients to book doctor consultations, attend online appointments, receive prescriptions, and manage their healthcare interactions digitally. The project faces frequent requirement changes. The Product Owner often modifies the backlog during sprint planning, which reduces team productivity and creates confusion among stakeholders.

Task

Analyze the Scrum roles and responsibilities, identify the causes of requirement instability, propose an improved backlog prioritization method, and recommend practical retrospective actions to improve team alignment and sprint effectiveness.

Deliverables

- Explanation of Scrum roles and responsibilities.
- Root-cause analysis of requirement instability.
- Improved product backlog prioritization plan.
- Retrospective improvement suggestions.

Reasoning

The situation represents common Agile project challenges such as requirement churn, weak backlog refinement, unclear priorities, and poor stakeholder alignment. Applying Scrum principles can make requirements more transparent, improve communication, and help the team deliver valuable telemedicine features in small, controlled increments.

Assignment Objective

The objective is to design a practical Scrum approach for the telemedicine application so that changing requirements are handled without continuously disrupting an active Sprint. The proposed approach emphasizes a clear Product Goal, a refined and ordered Product Backlog, stable Sprint Goals, and continuous improvement through Sprint Retrospectives.

1. SCRUM ROLES AND RESPONSIBILITIES

Scrum works effectively when the Product Owner, Scrum Master, and Developers have clear accountabilities. For the telemedicine application, each role should focus on its specific responsibility while collaborating closely with the others.

1.1 Product Owner

- **Owns and communicates the Product Goal** for the telemedicine application.
- **Manages and orders the Product Backlog** according to product value, risk, urgency, and dependencies.
- **Clarifies requirements** so Developers understand the expected outcome.
- **Engages stakeholders** such as patients, doctors, administrators, and business representatives.
- **Accepts or rejects outcomes** based on agreed requirements and product value.

Example: If appointment booking is more valuable for the next release than a new profile customization feature, the Product Owner should order appointment booking higher in the Product Backlog.

1.2 Scrum Master

- Helps the team understand and apply Scrum effectively.
- Facilitates Scrum events such as Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective.
- Removes or helps resolve impediments affecting the Developers.
- Protects the team from unnecessary disruption during the Sprint.
- Coaches stakeholders and the Scrum Team on effective Agile collaboration.

Example: If a stakeholder tries to insert an urgent feature directly into an active Sprint, the Scrum Master helps the team follow the agreed change process instead of allowing continuous interruption.

1.3 Developers

- Design, build, test, integrate, and maintain the telemedicine features.
- Create a usable and valuable Increment during each Sprint.
- Estimate and discuss the work required to achieve the Sprint Goal.
- Collaborate with the Product Owner when requirements need clarification.
- Maintain technical quality and identify implementation risks early.

1.4 Stakeholders

Doctors, patients, hospital administrators, support teams, and business representatives provide feedback and domain information. They should influence product direction through the Product Owner rather than continuously changing Developers' work during an active Sprint.

Role Alignment

Role	Main Responsibility	Telemedicine Example
Product Owner	Maximize product value and order backlog	Prioritize secure appointment booking
Scrum Master	Enable Scrum and remove impediments	Prevent uncontrolled Sprint disruption
Developers	Create the Increment	Build, test, and integrate consultation features
Stakeholders	Provide feedback and domain input	Review appointment and prescription workflows

2. ROOT-CAUSE ANALYSIS OF REQUIREMENT INSTABILITY

The main problem is not simply that requirements change. In Agile development, change is expected. The issue is that changes are entering the Sprint without sufficient refinement, prioritization, or stakeholder alignment.

Major Causes

- **Frequent Product Backlog changes:** The Product Owner modifies items during Sprint Planning, creating uncertainty.
- **Insufficient backlog refinement:** Requirements may not be sufficiently discussed, clarified, or split before planning.
- **Unclear acceptance criteria:** Developers may interpret the same requirement differently.
- **Multiple stakeholder requests:** Patients, doctors, management, and operations may have competing priorities.
- **Lack of a stable Sprint Goal:** The Sprint can become a collection of unrelated tasks rather than one valuable objective.
- **Weak communication:** Stakeholders may bypass the Product Owner and directly approach Developers.
- **Emergency requirements without a decision rule:** The team may not have a defined process for genuinely urgent changes.

Impact on the Project

- Developers spend time switching between tasks instead of completing planned work.
- Sprint commitments become less predictable.
- Testing and integration may be rushed.
- Stakeholders receive inconsistent information about what will be delivered.
- Team morale and productivity can decrease.
- Important telemedicine functions may be delayed.

Root-Cause Chain

Unclear stakeholder priorities → frequent backlog changes → unstable Sprint Planning → task switching → reduced productivity → delayed Increment → stakeholder confusion.

Recommended Control

The team should establish a regular Product Backlog Refinement activity before Sprint Planning. The Product Owner should gather stakeholder feedback, clarify requirements, define acceptance criteria, and order backlog items before they are considered for a Sprint.

Once a Sprint starts, the team should protect the Sprint Goal. A new request should normally enter the Product Backlog for future consideration. If a change makes the Sprint Goal obsolete, the Product Owner and Developers can discuss whether the Sprint should be changed or cancelled according to Scrum principles.

3. IMPROVED PRODUCT BACKLOG PRIORITIZATION PLAN

A clear prioritization method reduces arguments and helps the Product Owner explain why one feature is selected before another. A simple weighted approach can be used for this project.

Suggested Prioritization Factors

- **Customer Value:** How strongly the feature improves patient or doctor experience.
- **Business Value:** Contribution to adoption, revenue, or operational efficiency.
- **Risk/Security:** Importance of addressing safety, privacy, and security concerns early.

- **Urgency:** Whether the feature is time-sensitive or legally/operationally required.
- **Dependency:** Whether another important feature depends on this item.
- **Effort:** Development complexity and expected team capacity.

Example Ordered Backlog

Priority	Backlog Item	Reason
1	Secure doctor appointment booking	Core user value and essential workflow
2	Online consultation/video session	Enables the main telemedicine service
3	Digital prescription access	Important follow-up capability
4	Appointment reminders	Reduces missed consultations
5	Profile customization	Useful but lower immediate value

4. BACKLOG REFINEMENT AND SPRINT PLANNING IMPROVEMENT

Recommended Refinement Process

- Collect new requirements from stakeholders through the Product Owner.
- Group similar requests and remove duplicates.
- Clarify the user need and expected outcome.
- Write clear acceptance criteria.
- Split large features into smaller, testable Product Backlog Items.
- Discuss dependencies, risks, and technical considerations with Developers.
- Estimate effort and order the backlog using agreed prioritization factors.
- Select only sufficiently refined, high-priority items during Sprint Planning.

Change-Handling Rule During a Sprint

Situation	Recommended Action
Normal new feature request	Add it to the Product Backlog and prioritize it for a future Sprint.
Small clarification that does not change Sprint Goal	Clarify collaboratively and continue the Sprint.
Urgent issue affecting patient safety/security	Escalate immediately and assess impact with the Product Owner and Developers.
Change makes Sprint Goal obsolete	Discuss the Sprint impact; cancellation may be considered when appropriate.

5. SPRINT REVIEW AND STAKEHOLDER ALIGNMENT

The Sprint Review should be used to inspect the Increment and gather useful feedback. For the telemedicine project, representatives from patients, doctors, operations, and business teams can review completed functionality and discuss future needs.

- Demonstrate working features rather than only presenting documents.
- Record stakeholder feedback as backlog items or changes to existing items.
- Let the Product Owner decide backlog ordering based on product value.
- Avoid promising immediate implementation of every request.
- Communicate the next priorities clearly to reduce stakeholder confusion.

SCRUM ITERATION FOR THE TELEMEDICINE APP

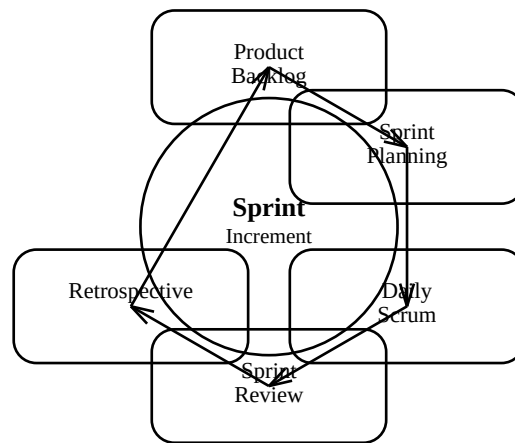


Figure 1: A simplified Scrum cycle for managing the telemedicine application.

6. RETROSPECTIVE IMPROVEMENT SUGGESTIONS

The Sprint Retrospective should focus on improving the team's process, quality, and collaboration. The team should identify a small number of actionable improvements rather than creating a long list that is difficult to implement.

Recommended Retrospective Discussion

Area	Observation	Action
Backlog	Items change too late	Hold regular refinement and define ready-to-discuss criteria
Communication	Stakeholders approach Developers directly	Route product requests through the Product Owner
Planning	Sprint scope becomes unstable	Agree on a clear Sprint Goal and protect it
Requirements	Acceptance criteria are unclear	Add testable acceptance criteria before selection
Team productivity	Frequent task switching	Limit mid-Sprint changes and focus on finishing work

Action Items for the Next Sprint

- Schedule one or more focused backlog refinement sessions before Sprint Planning.
- Create a simple Definition of Ready for items entering Sprint Planning.
- Use one clearly stated Sprint Goal.
- Maintain a visible ordered Product Backlog.
- Create a stakeholder communication channel through the Product Owner.
- Review whether the previous retrospective action items were completed.

7. EXPECTED IMPROVEMENTS

- Greater stability during Sprints.
- Less requirement churn and task switching.
- Better stakeholder understanding of priorities.
- Higher predictability of Sprint outcomes.
- Improved collaboration between Product Owner, Scrum Master, Developers, and stakeholders.
- Earlier identification of risks, dependencies, and unclear requirements.
- Better focus on valuable and testable telemedicine features.

8. CONCLUSION

The telemedicine application can benefit significantly from a disciplined Scrum approach. The Product Owner should maintain a clear Product Goal and an ordered Product Backlog, while the Scrum Master should protect the Scrum process and help remove impediments. Developers should focus on creating a usable Increment and maintaining technical quality.

The requirement instability in the scenario is mainly caused by frequent late changes, inadequate backlog refinement, unclear acceptance criteria, and weak stakeholder alignment. A structured prioritization method, regular refinement, clear Sprint Goals, and controlled change handling can reduce these problems. Finally, Sprint Retrospectives should produce small, measurable actions that continuously improve the team's way of working.

Overall Result: By applying Scrum consistently, the telemedicine team can manage changing requirements without sacrificing Sprint focus, improve stakeholder communication, and deliver valuable features in a predictable and continuously improving manner.